

Governing Interruptions After Deterministic Deterioration Scoring: A Retrospective Evaluation on PhysioNet Challenge 2019

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Abstract

Objectives: A deterioration-score threshold may produce repeated, poorly timed interruptions. We evaluated a frozen dual-lane governance policy for partial SOFA, compared it with threshold-only partial SOFA and hourly bedside scores, and attributed its effect to individual mechanisms. **Materials and Methods:** We selected governance on PhysioNet Challenge 2019 setA ($n = 20,336$), froze it, and evaluated setB ($n = 20,000$) once. Detection required an emission within $[\text{label_start} - 12\text{ h}, +6\text{ h}]$; burden counted interruptive emissions. Co-primary outcomes were detection/burden metrics and normalized Challenge utility. Confidence intervals used 1,000 stay-level bootstrap replicates.

Results: Governed partial SOFA matched threshold-only sensitivity (79.5%, 95% CI 77.0–81.8). The interruptive lane emitted 13.2% as many alerts, detected 34.0% of positive stays, produced 106.1 emissions per detected positive stay (95% CI 96.2–116.6), and had 10.7% stay-level positive predictive value. Normalized utility was -0.111 , -0.053 , and 0.041 for threshold-only, governed, and interruptive outputs, respectively. Removing the 90-minute refractory interval doubled volume; disabling the page gate increased interruptive sensitivity to 44.6%.

Discussion: A passive lane structurally preserved threshold-crossing detection while refractory and page-gate rules reduced interruptive emissions. Residual burden remained substantial.

Conclusion: This single-resource, partial-score evaluation supports further governance research but does not establish diagnosis, clinical benefit, external validity, or readiness for patient care.

Lay summary

Hospital early-warning systems may interrupt clinicians so often that important warnings compete with routine ones. We studied whether the decision to interrupt can be separated from the calculation of a patient’s risk score. Using public intensive-care records, we chose the interruption rules on one part of the dataset, locked them, and tested them once on the other part. A passive notification lane identified the same proportion of labeled patients as the raw score, while a stricter urgent lane produced about one eighth as many notifications. A quiet period after each notification and a rule reserving urgent notifications for stronger trajectories produced most of this reduction. The urgent lane still generated 1.30 notifications per patient-day and detected only about one third of labeled patients, so it is not ready for clinical use. The dataset’s label partly depends on the same organ-dysfunction concept as the score, and the available data support only an incomplete SOFA score. The findings therefore concern alert routing, not diagnosis, treatment benefit, or patient outcomes.

1 Background and Significance

Electronic clinical decision support can generate technically correct but operationally weak interruptions. Repeated exposure can reduce responsiveness, and interruption cost depends on relevance and workflow fit rather than alert count alone.[1, 2] Randomized sepsis-alert evaluations have also produced mixed clinical results, showing that model discrimination does not by itself determine the value of an alerting workflow.[3]

Retrospective deterioration studies commonly treat each positive score as an alert. This collapses scoring, routing, and clinician interruption into one event. We instead separate them. A deterministic scorer produces an auditable signal; a shared governance layer then suppresses repetition or routes the signal to a passive watch or interruptive lane. An optional language model may explain an already-fired alert but cannot create, suppress, escalate, or alter the score.

We evaluated this separation using the PhysioNet/Computing in Cardiology Challenge 2019 public resource, which provides hourly intensive-care records and an onset-aligned `SepsisLabel`. [4, 5] Because the label incorporates a SOFA change, detection by a SOFA-derived score is not an independent diagnostic validation. The estimand is the change in retrospective alert-stream behavior caused by governance after the same score.

2 Objectives

The primary objective was to determine whether a policy selected on setA and frozen before setB replay reduced page-eligible emissions while retaining threshold-only partial-SOFA detection. Secondary objectives were to quantify Challenge utility, compare the operating point with hourly SIRS, incomplete NEWS2, and partial qSOFA, and identify which governance mechanisms produced the observed changes.

3 Materials and Methods

3.1 System boundary

The Curie research prototype projects clinical events onto a canonical envelope, transports them through Kafka, and applies versioned deterministic rules in Apache Flink. Each score includes evi-

dence identifiers, missing components, rule version, and rule-content hash. The retrospective harness used the same scoring and governance semantics but did not simulate delivery, acknowledgement, or treatment (Figure 1).

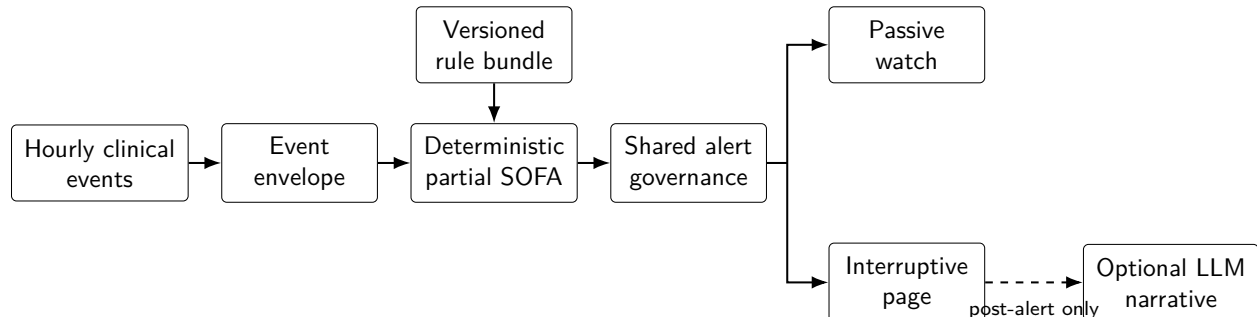


Figure 1: System boundary. The language-model component is outside the alert path. The evaluation ends at an offline routing decision.

Alt text. Left-to-right system diagram. Hourly clinical events flow into an event envelope and then a deterministic partial-SOFA scorer supplied by a versioned rule bundle. The score flows to shared alert governance, which branches upward to a passive watch lane and downward to an interruptive page lane. A dashed arrow from the page lane reaches an optional language-model narrative labeled post-alert only, emphasizing that the language model is outside the alert path.

3.2 Dataset and split

We used PhysioNet Challenge 2019 version 1.0.0 under CC BY 4.0. SetA contained 20,336 stays and setB 20,000 stays.[4, 5] SetA was used for selection and setB as a locked retrospective holdout. SetB was not used to tune or reselect the policy. This was neither an official Challenge submission nor evaluation on the hidden Challenge test set, and setB is not an external hospital cohort.

3.3 Partial SOFA and threshold-only output

SOFA summarizes dysfunction in six organ systems.[6] Challenge records permitted an incomplete proxy reconstruction from oxygenation, mean arterial pressure, platelets, bilirubin, and creatinine. GCS, urine-output windows, vasopressor doses, and reliable ventilation status were unavailable. Values were carried forward causally within each stay in the frozen replay; future rows were never used. The respiratory proxy divided O2Sat (or SaO2 when O2Sat was absent) by FiO_2 and applied the numeric P/F bands; this is not a standard SOFA respiratory component or a validated S/F conversion. Other available components used conventional SOFA cut points (Supplementary

Table S1), missing components were explicit, and at least two components were required. The threshold-only lane emitted at each hour with partial SOFA ≥ 2 . Thus neither a missing component nor an unavailable organ system was assigned a reassuring zero.

3.4 Governance and operating-point freeze

The governed watch path required one qualifying crossing and no persistence or baseline-delta gate. A 90-minute refractory interval suppressed repeated emissions. The interruptive page gate additionally required two crossings, 30 minutes at page-level severity, a component-score increase of at least one, and at least two positive components. Challenge rows are hourly, so the 30-minute and two-crossing settings may be observationally redundant.

The original setA sweep evaluated 23 candidates under a legacy grace-6 detection definition. A candidate passed the sensitivity objective if governed sensitivity was at least naive sensitivity minus 0.10 or at least 0.70, and passed the burden objective if its interruptive-to-naive emission ratio was at most 0.25. Candidates were ranked by passing both objectives, lower burden ratio, lower emissions per detected positive stay, and higher governed sensitivity. The selected candidate, `grid_p0_r90_b0`, and complete selection grid appear in Supplementary Table S2. The configuration and resolved rule bundle were frozen before setB evaluation.

3.5 Comparators

We independently scored hourly SIRS ≥ 2 using temperature, heart rate, respiratory rate, and white-cell count;[7] NEWS2 ≥ 5 with the unavailable consciousness/AVPU item assigned zero;[8] and partial qSOFA ≥ 2 using respiratory rate ≥ 22 and systolic blood pressure ≤ 100 , omitting GCS.[9] These threshold-only streams had no refractory or page gate. The comparison is descriptive because missing items disadvantage NEWS2 and qSOFA and none received matched governance.

3.6 Outcomes and statistical analysis

For positive stays, `SepsisLabel` starts approximately six hours before the Challenge clinical-onset definition.[4] We call the first positive hour `label_start`, not bedside onset. Primary-window detection required at least one emission in `[label_start - 12 h, label_start + 6 h]`. Detection used any governed emission; burden used interruptive emissions.

We report sensitivity, total emissions, interruptive-to-naive emission ratio, emissions per detected positive stay, alerts per patient-day, and stay-level positive predictive value (PPV), defined as labeled positive stays detected in the window divided by all stays with any emission. The PPV denominator is not restricted to a matched onset window in negative stays. Lead time is from the first in-window governed emission to `label_start`; negative values are after label start. Confidence intervals are percentile intervals from 1,000 stay-level bootstrap replicates (seed 42).

Normalized PhysioNet Challenge utility was co-primary and used the official hourly weighting implementation. Positive predictions were emission hours only, not latched states; consequently this is neither a leaderboard score nor a utility estimate for delivered clinical pages. Drop-one ablations modified one frozen mechanism at a time without reselection. Alternative timing definitions were secondary analyses (Supplementary Table S3).

4 Results

4.1 Selection and holdout cohorts

SetA contained 1,790 label-positive stays. Under the original legacy grace-6 selection estimand, the winner had 88.4% governed sensitivity, 37.7% interruptive sensitivity, and a 0.123 interruptive-emission ratio. Applying the later primary-window definition to the same winner yielded 87.6% governed and 32.5% interruptive sensitivity (Supplementary Table S2); the distinction prevents post hoc substitution of estimands.

SetB contained 1,142 positive and 18,858 negative stays and 31,749.8 patient-days. Threshold-only, governed, and interruptive outputs comprised 311,797, 159,933, and 41,158 emissions, respectively. Primary results are in Table 1.

Governed and threshold-only sensitivity were both 79.5% (908/1,142), a result expected from a watch path that required one crossing and had no persistence or baseline gate. Interruptive sensitivity was 34.0% (388/1,142). The interruptive lane emitted 13.2% as many alerts as threshold-only scoring and 1.30 alerts per patient-day, but still required 106.1 emissions per detected positive stay. Among negative stays, 3,238/18,858 (17.2%) had at least one interruptive emission; their mean was 1.70 emissions per stay and median was 0 (IQR 0–0). Stay-level interruptive PPV was 10.7% (95% CI 9.7–11.7).

Table 1: Primary setB holdout results under the prespecified $-12/+6$ -hour window. CI, confidence interval; PPV, positive predictive value; N/G/I, threshold-only naive/governed/interruptive.

Metric	Result	95% CI	Interpretation
Governed sensitivity (%)	79.5	77.0–81.8	Any governed emission in-window
Interruptive sensitivity (%)	34.0	31.3–36.8	Interruptive emission in-window
Interruptive emissions / detected positive stay	106.1	96.2–116.6	Emission-based burden; not episode pages
Interruptive / naive emissions	0.132	0.127–0.137	Page-eligible vs threshold-only SOFA
Interruptive alerts / patient-day	1.30	—	All setB patient-time
Interruptive stay-level PPV (%)	10.7	9.7–11.7	Any emit during stay; not onset-window PPV
Mean in-window lead (h)	5.97	5.54–6.39	First governed emission to <code>label_start</code>
Challenge utility (N/G/I)	-0.111 / -0.053 / 0.041	—	Emit-hour predictions; co-primary

Normalized Challenge utility was -0.111 for threshold-only emissions, -0.053 for any governed emission, and 0.041 for the interruptive lane. Thus the most selective stream improved this retrospective utility relative to inaction, while the other two remained below it.

Mean first-emission lead was 5.97 hours (95% CI 5.54–6.39), but the distribution was asymmetric: the 25th percentile was -1 hour, median 11 hours, and 75th percentile 12 hours. A mean alone obscures that at least one quarter of detected positive stays first emitted after `label_start`.

4.2 Comparators and ablations

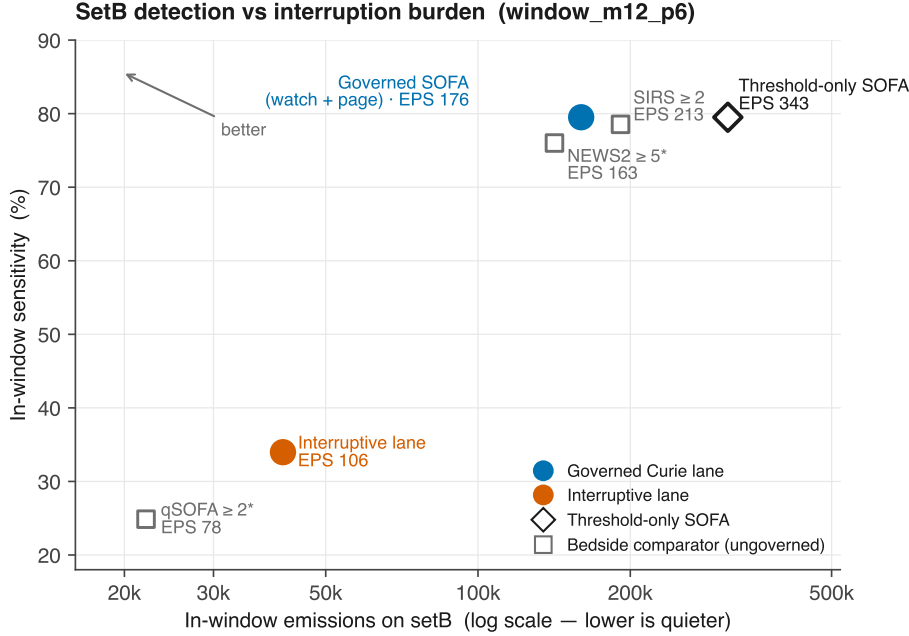
Incomplete NEWS2 and SIRS had 76.0% and 78.5% sensitivity but generated ungoverned hourly streams requiring 163–213 emissions per detected positive stay (Table 2; Figure 2). Partial qSOFA was quieter but detected 24.9%. These results do not establish clinical superiority because the scores were incomplete and their routing policies were unmatched.

Removing the refractory interval increased governed volume to threshold-only volume. Disabling the page gate increased interruptive sensitivity from 34.0% to 44.6% and its emission ratio from 0.132 to 0.457 (Table 3). Disabling baseline had no effect because it was already off. Zeroing page persistence or reducing page crossings had no effect on hourly rows.

Of 1,142 positive setB stays, 234 were governed false negatives in the primary window. All were

Table 2: SetB threshold-only comparators and Curie lanes under the primary window. Missing NEWS2 consciousness and qSOFA GCS components limit comparison.

Policy	Sensitivity (%)	Emissions	Emit./detected positive stay
SIRS ≥ 2 (Temp, HR, RR, WBC)	78.5	191294	213.3
NEWS2 ≥ 5 (consciousness omitted)	76.0	141629	163.2
Partial qSOFA ≥ 2 (RR and SBP; GCS omitted)	24.9	22068	77.7
Threshold-only partial SOFA	79.5	311797	343.4
Governed partial SOFA (any emit)	79.5	159933	176.1
Interruptive governed lane	34.0	41158	106.1



*NEWS2 omits consciousness/AVPU; qSOFA omits GCS (absent in Challenge 2019). EPS = emissions per in-window detected positive stay. Not a claim of clinical superiority.

Figure 2: SetB sensitivity versus emission burden. Markers represent unmatched threshold-only comparator streams and Curie output lanes.

Alt text. Scatter plot of setB sensitivity against emissions per detected positive stay for hourly SIRS, incomplete NEWS2, partial qSOFA, threshold-only partial SOFA, any governed partial-SOFA emission, and the interruptive governed lane. The interruptive lane has lower burden and lower sensitivity than any governed emission; comparator points are descriptive because missing score components and routing policies are unmatched.

Table 3: Drop-one setB ablation of the frozen winner. No variant was reselected.

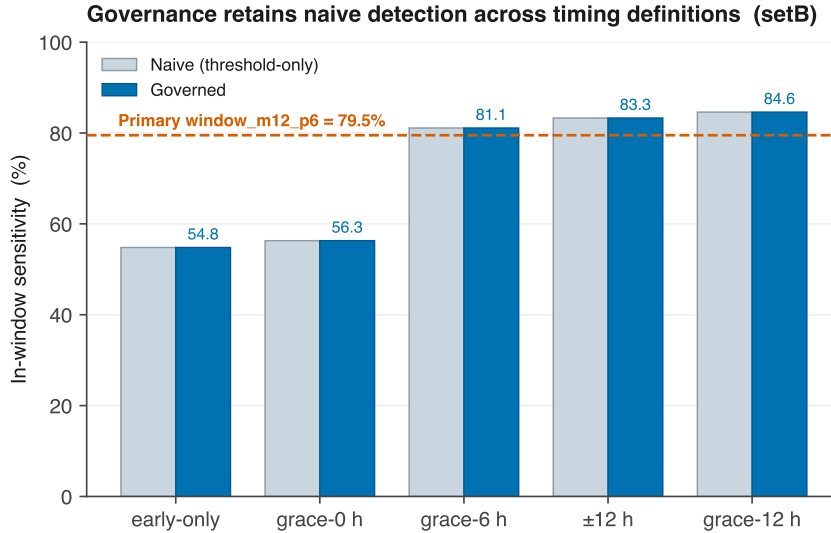
Variant	Gov. sens. (%)	Int. sens. (%)	Int. reduction	Int. EPS	Note
primary_operating_point	79.5	34.0	0.132	106.1	Frozen winner
drop_baseline	79.5	34.0	0.132	106.1	Null (already off)
drop_persistence	79.5	34.0	0.132	106.1	No change on hourly rows
drop_crossings	79.5	34.0	0.132	106.1	No change on hourly rows
drop_refractory	79.5	36.2	0.264	199.6	Watch volume = naive
drop_page_gate	79.5	44.6	0.208	127.5	Page gate disabled

attributed to never crossing the partial-SOFA threshold in the window rather than governance suppression (Table 4). This finding is structural for the selected watch path, not evidence that governance can generally preserve detection.

Table 4: Aggregate attribution of governed false negatives on setB. No stay identifiers are included.

Reason	Count	Rate (%)
Never crossed naive SOFA threshold in-window	141	60.3
Governed emit only outside the window	58	24.8
Never scoreable (missing inputs)	35	15.0

Alternative timing definitions changed absolute sensitivity but preserved equal threshold-only and governed sensitivity for this configuration (Supplementary Table S3; Figure 3). The 81.1% grace-6 setB result is secondary and does not replace the 79.5% primary-window estimate.



Secondary sensitivity analyses; naive and governed sensitivity are equal for the frozen winner in every definition. The 81.1% grace-6 h value is secondary, not the primary result.

Figure 3: Threshold-only and governed setB sensitivity under secondary timing definitions. The dashed line marks primary governed sensitivity (79.5%).

Alt text. Grouped comparison of threshold-only and governed sensitivity across secondary timing definitions on setB. The paired values are equal for every displayed definition. A dashed horizontal reference marks the 79.5 percent governed sensitivity under the primary minus-12 to plus-6-hour window.

5 Discussion

The frozen policy reduced page-eligible emissions to 13.2% of threshold-only partial-SOFA volume. The refractory interval reduced repeated watch emissions, and the page gate traded interruptive sensitivity for lower burden. Challenge utility moved from negative for threshold-only and all governed emissions to slightly positive for the interruptive stream. These findings support studying alert routing as a versioned intervention distinct from the score.

Equal threshold-only and governed sensitivity is not an independent performance gain. With one required watch crossing, no persistence, no baseline gate, and no usable suppression flags in this dataset, any threshold crossing produced a governed watch unless refractory timing moved it outside the analysis window. The equality therefore mainly verifies that this routing configuration preserved crossings. The empirical trade-off lies in the interruptive page gate.

Residual burden remained substantial: 1.30 interruptive emissions per patient-day, 106 emissions per detected positive stay, and at least one interruptive emission in 17.2% of negative stays. Emission counts may overstate or understate real interruption cost because replay had no episode arbitration, delivery failures, acknowledgements, escalation, or clinician response. A usable policy would need workflow-specific grouping and prospective burden measurement.

The label-score relationship further limits interpretation. Challenge `SepsisLabel` uses suspected infection and an acute SOFA increase, so a SOFA-derived detector benefits from incorporation.[4] This does not invalidate the within-score governance comparison, but it prevents treating sensitivity as diagnostic validation. Comparator results are also descriptive because NEWS2 and qSOFA were incomplete and ungoverned.

6 Limitations

First, this was one public resource with a competition proxy label; setB is not external validation. Second, partial SOFA omitted major organ-system inputs, used a nonstandard S/F respiratory proxy with P/F numeric bands, and used causal carry-forward, which may not match bedside availability or documentation latency. Third, the 23-candidate search and selection objectives were specified within setA but were not registered prospectively. Fourth, bootstrap intervals quantify resampling uncertainty within setB, not site or temporal transportability. Fifth, outputs were offline

emissions rather than delivered clinical interruptions. We did not assess calibration, subgroups, fairness, clinician response, treatment, or patient outcomes.

7 Conclusion

In a frozen setA-to-setB replay, a dual-lane policy preserved partial-SOFA threshold crossings on a passive lane while refractory and page-gate rules reduced interruptive emissions. The improvement in Challenge utility and lower volume justify further governance research, but residual burden, a partial score, incorporation-prone labels, and absent external or prospective validation preclude clinical-use claims.

Author contributions

Satya Venkata Ranga Janaki Sriram Mentey: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review and editing, and Project administration.

Data availability

PhysioNet Challenge 2019 version 1.0.0 is available at <https://doi.org/10.13026/v64v-d857> under CC BY 4.0. Patient-level inputs and replay outputs are not redistributed. Frozen aggregate sidecars, rule bundles, hashes, table-generation code, and the reproducibility manifest are at <https://github.com/sreeram843/curie-prediction-pipeline>. Manuscript values are pinned to commit 1d8a1117151619e2d34822e365d17a00e30e5233; later scorer revisions are outside this frozen study. Tables regenerate with `make paper-tables`.

Ethics approval

This analysis used only the publicly available, de-identified PhysioNet Challenge 2019 dataset and involved no new data collection or participant contact. The author has no institutional affiliation that requires ethics-board review of this secondary analysis; institutional ethics review was not

required.

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Competing interests

The author declares no competing interests.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used OpenAI Codex / GPT-5 and related assistants for software implementation support, manuscript organization, language editing, code review, and L^AT_EX formatting. The author reviewed and edited the output as needed and takes full responsibility for the content of the published article. These tools are not authors. Reported analyses use deterministic scoring and rule-based governance; an optional language-model narrative component is outside the alert path and is not part of any reported metric.

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